

Applying Exponent Laws

MA.8.NSO.1.3

I can rewrite a rational base with a negative exponent using a positive exponent, and apply several laws to generate an equivalent expression.

Yesterday: a negative exponent means reciprocal.

Today the base is a fraction — so the reciprocal flips it.

Warm-Up

On your own. Four minutes. No notes.

1. Find the value: 2^{-3}
2. Rewrite with a positive exponent: $\frac{1}{7^{-2}}$
3. Which is larger, 4^3 or 3^4 ?
4. Write the reciprocal of $-\frac{2}{5}$

Warm-Up

Answers.

1. Find the value: 2^{-3} $\frac{1}{8}$
2. Rewrite with a positive exponent: $\frac{1}{7^{-2}}$ 7^2
3. Which is larger, 4^3 or 3^4 ? 3^4 **(81 > 64)**
4. Write the reciprocal of $-\frac{2}{5}$ $-\frac{5}{2}$

Notes

Copy the table.

I. The pattern with a fraction base

Expression	Expanded	Value
$(2/5)^2$	$2/5 \cdot 2/5$	$4/25$
$(2/5)^1$	$2/5$	$2/5$
$(2/5)^0$	1	1
$(2/5)^{-1}$	$5/2$	$5/2$
$(2/5)^{-2}$	$5/2 \cdot 5/2$	$25/4$

Notes

II. Flip the base, make the exponent positive

$$\left(\frac{a}{b}\right)^{-n} = \left(\frac{b}{a}\right)^n \quad (a, b \neq 0)$$

$$\left(\frac{2}{5}\right)^{-2} = \left(\frac{5}{2}\right)^2 = \frac{25}{4} \qquad \left(-\frac{2}{3}\right)^{-5} = \left(-\frac{3}{2}\right)^5$$

The sign stays with the fraction. Flip the numbers, not the sign.

A whole number is a fraction over 1: $4^{-3} = \left(\frac{4}{1}\right)^{-3} = \left(\frac{1}{4}\right)^3$

Notes

III. Several laws at once — the plan is still yours

$$\left(\frac{2.5^5}{2.5^3}\right)^{-1} = (2.5^2)^{-1} = 2.5^{-2} = \left(\frac{5}{2}\right)^{-2} = \left(\frac{2}{5}\right)^2 = \frac{4}{25}$$

Any correct order of laws gives the same value. Finish with a positive exponent.

Example 1

Flip, then evaluate.

$$\left(-\frac{7}{8}\right)^{-2}$$

Rewrite with a positive exponent, then find the value.

Example 1

Worked.

$$\left(-\frac{7}{8}\right)^{-2} = \left(-\frac{8}{7}\right)^2 \quad \text{flip the base; the sign stays}$$

$$\left(-\frac{8}{7}\right)^2 = \frac{64}{49} \quad \text{even power: positive}$$

Answer: 64/49

Your Turn

Same steps, your numbers. Boards up when done.

$$\left(-\frac{5}{2}\right)^{-3}$$

Your Turn

Answer.

$$\left(-\frac{5}{2}\right)^{-3}$$

$(-2/5)^3$ — odd power, negative

Answer: $-\frac{8}{125}$

Example 2

Two laws, then a flip.

$$\left(\frac{5}{8}\right)^{-2} \cdot \left(\frac{5}{8}\right)^4$$

Write an equivalent expression with one base and a positive exponent.

Example 2

Worked.

$$\left(\frac{5}{8}\right)^{-2} \cdot \left(\frac{5}{8}\right)^4 = \left(\frac{5}{8}\right)^{-2+4} = \left(\frac{5}{8}\right)^2 \quad \text{product of powers}$$

$$\left(\frac{5}{8}\right)^2 = \frac{25}{64} \quad \text{already positive: no flip}$$

$$\textbf{Answer: } \left(\frac{5}{8}\right)^2 = \frac{25}{64}$$

Your Turn

Same steps, your numbers. Boards up when done.

$$\left(-\frac{5}{4}\right)^{-7} \cdot \left(-\frac{5}{4}\right)^{-3}$$

Your Turn

Answer.

$$\left(-\frac{5}{4}\right)^{-7} \cdot \left(-\frac{5}{4}\right)^{-3}$$

exponent -10 , then flip: $(-4/5)^{10}$

Answer: $\left(-\frac{4}{5}\right)^{10}$

Whiteboards • Question 1 of 9

Boards up on three.

$$\left(\frac{2}{3}\right)^{-4}$$

Answer it.

Rewrite with a positive exponent.

Whiteboards • Question 1 of 9

Answer.

$$\left(\frac{2}{3}\right)^{-4}$$

flip the base

Answer: $\left(\frac{3}{2}\right)^4$

Whiteboards • Question 2 of 9

Boards up on three.

$$\left(\frac{3}{-5}\right)^{-9}$$

Answer it.

Rewrite with a positive exponent.

Whiteboards • Question 2 of 9

Answer.

$$\left(\frac{3}{-5}\right)^{-9}$$

flip the numbers; the sign stays in the fraction

Answer: $\left(\frac{-5}{3}\right)^9$

Whiteboards • Question 3 of 9

Boards up on three.

$$\left(-\frac{8}{9}\right)^{-2}$$

Answer it.

Find the value.

Whiteboards • Question 3 of 9

Answer.

$$\left(-\frac{8}{9}\right)^{-2}$$

$$(-9/8)^2 = 81/64$$

Answer: $\frac{81}{64}$

Whiteboards • Question 4 of 9

Boards up on three.

$$\left(\frac{21}{4}\right)^{-5}$$

Answer it.

Rewrite with a positive exponent.

Whiteboards • Question 4 of 9

Answer.

$$\left(\frac{21}{4}\right)^{-5}$$

flip

Answer: $\left(\frac{4}{21}\right)^5$

Whiteboards • Question 5 of 9

Boards up on three.

$$\left(-\frac{7}{8}\right)^{-2}$$

Which expression is equivalent?

A. $(-8/7)^2$

B. $(8/7)^2$

C. $-(8/7)^2$

D. $(-7/8)^2$

Answer it.

Whiteboards • Question 5 of 9

Answer.

$$\left(-\frac{7}{8}\right)^{-2}$$

Which expression is equivalent?

A. $(-8/7)^2$

B. $(8/7)^2$

C. $-(8/7)^2$

D. $(-7/8)^2$

Answer: A — $(-8/7)^2$

Whiteboards • Question 6 of 9

Boards up on three.

$$\left(\frac{5}{8}\right)^{-2} \div \left(\frac{5}{8}\right)^{-4}$$

Which expression is equivalent?

A. $(5/8)^2$

B. $(8/5)^2$

C. $(5/8)^{-6}$

D. $(5/8)^8$

Answer it.

Whiteboards • Question 6 of 9

Answer.

$$\left(\frac{5}{8}\right)^{-2} \div \left(\frac{5}{8}\right)^{-4}$$

Which expression is equivalent?

A. $(5/8)^2$

B. $(8/5)^2$

C. $(5/8)^{-6}$

D. $(5/8)^8$

Answer: A — $(5/8)^2$

Whiteboards • Question 7 of 9

Boards up on three.

$$\left(\frac{2}{5}\right)^{-1} \cdot \left(\frac{5}{2}\right)^1$$

Answer it.

Find the value.

Whiteboards • Question 7 of 9

Answer.

$$\left(\frac{2}{5}\right)^{-1} \cdot \left(\frac{5}{2}\right)^1$$

$$(5/2)^1 \cdot (5/2)^1 = (5/2)^2$$

$$\textbf{Answer: } \frac{25}{4}$$

Whiteboards • Question 8 of 9

Boards up on three.

$$\left[\left(\frac{2}{3}\right)^5 \cdot \left(\frac{2}{3}\right)^{-2}\right]^3$$

Answer it.

Write it with one base and one positive exponent.

Whiteboards • Question 8 of 9

Answer.

$$\left[\left(\frac{2}{3}\right)^5 \cdot \left(\frac{2}{3}\right)^{-2}\right]^3$$

$(2/3)^3$ inside, then power of a power

Answer: $\left(\frac{2}{3}\right)^9$

Whiteboards • Question 9 of 9

Take your time. Boards up when you have written it.

Marco says $\left(-\frac{2}{3}\right)^{-5}$ can be rewritten as $\left(\frac{2}{3}\right)^5$

because 'two negatives make a positive.' Is he right?

Write your answer in sentences.

Say what the negative exponent does, and what it does not do. Give the correct rewrite.

Whiteboards • Question 9 of 9

Answer.

Marco says $\left(-\frac{2}{3}\right)^{-5}$ can be rewritten as $\left(\frac{2}{3}\right)^5$

because 'two negatives make a positive.' Is he right?

the exponent flips the fraction; it never touches the sign

Answer: No — $(-2/3)^{-5} = (-3/2)^5$

IXL

Last five minutes.

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Stop at a SmartScore of 67 on each skill. Whatever you do not finish is tonight's practice.

Today's skills

Identify equivalent expressions involving exponents I — VLM